Unit 08: HUMAN STUDIES

Scientific Research

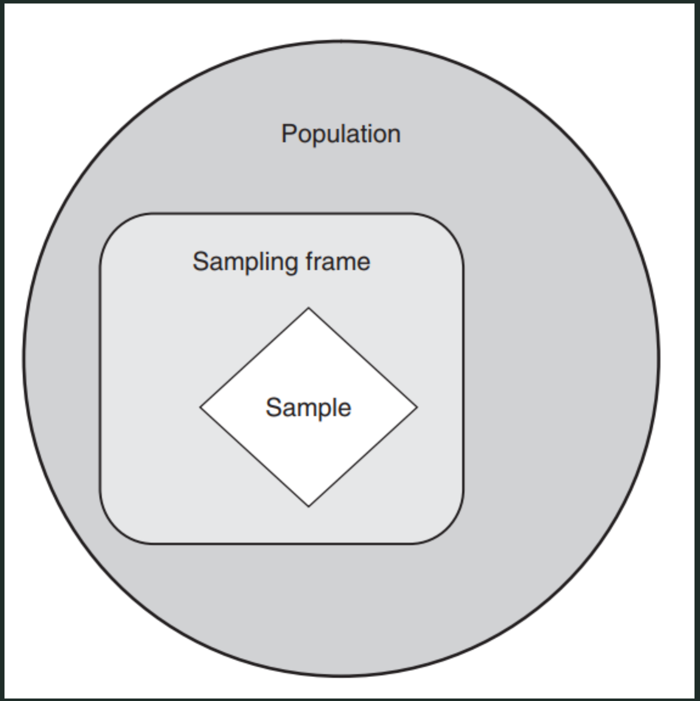
1. COLLECTING PRIMARY DATA

SAMPLING

- Before we explore the different methods of data collection, we need to consider the issue of selecting who to question or what to examine when faced with a large number of cases.
- If you want to get information about a large group of individual people or things, for example, students or cars, it is normally **impossible to get all of them** to answer your questions or to examine all the things – it would take much **too long** and be **far too expensive**.
- The solution is to just ask or examine some of them and hope that the data you get are **representative** (or typical) of all the rest.
- This process of selecting just a small group of cases from out of a large group is called **sampling**.

SAMPLING

- When doing a survey, the question inevitably arises: how similar are characteristics of the sample that are chosen for the survey to those of the whole population?
- When we talk about **population** in research, it does not necessarily mean a number of people, it is a collective term used to describe the total quantity of things (or cases) of the type which are the subject of your study.
- Within this population, there will probably be only certain groups that will be of interest to your study, for instance, of all school buildings only those in cities, or of all limited companies, only small to medium sized companies. This selected category is your **sampling frame**.



SAMPLING

- Difficulties are encountered if the characteristics of the population are not known, or if it is not possible to reach sectors of it. Non-representative samples cannot be used to make accurate generalizations about the population.
- Populations can have the following characteristics:
  - **homogeneous** – all cases are similar , e.g. bottles of beer on a production line;
  - **stratified** – contain strata or layers , e.g. people with different levels of income: low, medium, high;
  - **proportional stratified** – contains strata of known proportions e.g. percentages of different nationalities of students in a university;
  - **grouped by type** – contains distinctive groups, e.g. of apartment buildings – towers, slabs, villas, tenement blocks;
  - **grouped by location** – different groups according to where they are e.g. animals in different habitats – desert, equatorial forest, savannah, tundra.

SAMPLING

- It is generally accepted that conclusions reached from the study of a large sample are more convincing than those from a small one.
- However, *the preference for a large sample must be balanced against the practicalities of the research resources*, i.e. cost, time and effort.
  - If the population is very homogeneous, and the study is not very detailed, then a small sample will give a fairly representative view of the whole.
  - If statistical tests are to be used to analyze the data, there are usually minimum sample sizes specified from which any significant results can be obtained.
    - The size of the sample also should be in direct relationship to the number of variables to be studied.

SAMPLING

- No sample will be exactly representative of a population.
- If different samples, using identical methods, are taken from the same population, there are bound to be differences in the mean (average) values of each sample owing to the chance selection of different individuals.
- The measured difference between the mean value of a sample and that of the population is called the **sampling error**, which will lead to bias in the results.
- **Bias** is the unwanted distortion of the results of a survey due to parts of the population being more strongly represented than others.

PROBABILITY SAMPLING

- Probability sampling techniques give the most reliable representation of the whole population.
- This is based on **using random methods** to select the sample.
- Populations are not always quite as uniform or one-dimensional as, say, a particular type of component in a production run, so simple random selection methods are not always appropriate.
- The select procedure should aim to guarantee that each element (person, group, class, type etc.) has **an equal chance of being selected** and that every possible combination of the elements also has an equal chance of being selected.

PROBABILITY SAMPLING

- Therefore, the first question to be asked is about **the nature of the population**: is it homogeneous or are there distinctly different classes of cases within it, and if so, how are they distributed within the population (e.g. are they grouped in different locations, found at different levels in a hierarchy or are they all mixed up together)?
- Specific techniques are used for selecting representative samples from populations of the different characteristics, such as
  - simple random sampling,
  - stratified sampling,
  - cluster sampling etc.

SIMPLE RANDOM SAMPLING (SRS)

- The most commonly used method of selecting a probability sample.
- In line with the definition of randomization, whereby each element in the population is given an equal and independent chance of selection, a simple random sample is selected by the procedure:
    1. Identify by a number all elements or sampling units in the population.
    2. Decide on the sample size *n*.
    3. Select *n individuals* using the table of random numbers or a computer program.
- To illustrate, let us take the example of a class. There are 80 students in the class, and so the first step is to identify each student by a number from 1 to 80. Suppose you decide to select a sample of 20 using the simple random sampling technique. Use a computer program to select the 20 students. These 20 students become the basis of your enquiry.

PROPORTIONATE STRATIFIED RANDOM SAMPLING

- In stratified random sampling the researcher attempts to **stratify the population** in such a way that the population within a stratum is homogeneous with respect to the characteristic on the basis of which it is being stratified.
- It is important that the characteristics chosen as the basis of stratification are clearly identifiable in the study population.
  - For example, it is much easier to stratify a population on the basis of gender than on the basis of age, income or attitude.
- It is also important for the characteristic that becomes the basis of stratification to be related to the main variable that you are exploring.
- Once the sampling population has been separated into non-overlapping groups, you select the required number of elements from each stratum, using the simple random sampling technique.
- In proportionate stratified sampling, the number of elements from each stratum in relation to its proportion in the total population is selected.

CLUSTER SAMPLING

- Simple random and stratified sampling techniques are based on a researcher's ability to identify each element in a population.
- It is easy to do this if the total sampling population is small.
- However, if the population is large, as in the case of a city, state or country, it becomes difficult and expensive to identify each sampling unit.
- In such cases the use of cluster sampling is more appropriate.

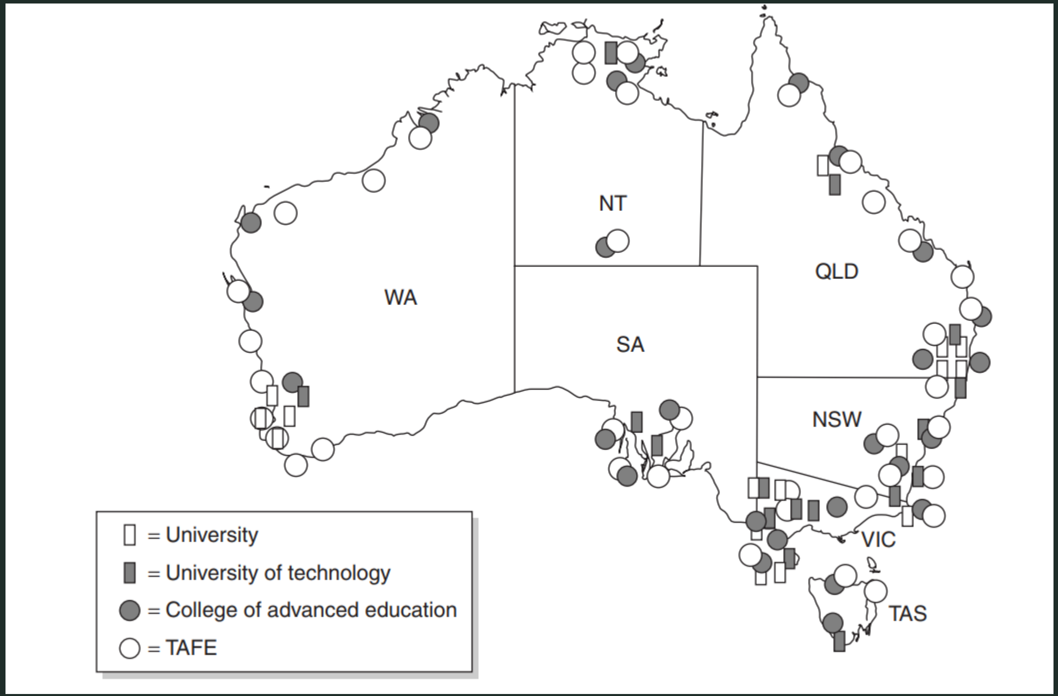
CLUSTER SAMPLING

- Cluster sampling is based on the ability of the researcher to divide the sampling population into groups (based upon visible or easily identifiable characteristics), called clusters, and then to select elements within each cluster, using the SRS technique.
- Clusters can be formed on the basis of geographical proximity or a common characteristic that has a correlation with the main variable of the study (as in stratified sampling).
- Depending on the level of clustering, sometimes sampling may be done at different levels.
- These levels constitute the different stages (single, double or multiple) of clustering.

CLUSTER SAMPLING
Example

- Imagine you want to investigate the attitude of post-secondary students in Australia towards problems in higher education in the country. Higher education institutions are in every state and territory of Australia. In addition, there are different types of institutions, for example universities, universities of technology, colleges of advanced education and colleges of technical and further education (TAFE).

- Within each institution various courses are offered at both undergraduate and postgraduate levels. each academic course could take three to four years. You can imagine the magnitude of the task. In such situations cluster sampling is extremely useful in selecting a random sample.



CLUSTER SAMPLING
Example

The first level of cluster sampling could be at the state or territory level. Clusters could be grouped according to similar characteristics that ensure their comparability in terms of student population. If this is not easy, you may decide to select all the states and territories and then select a sample at the institutional level. For example, with a simple random technique, one institution from each category within each state could be selected (one university, one university of technology and one TAFE college). This is based upon the assumption that institutions within a category are fairly similar with regards to student profile. Then, within an institution on a random basis, one or more academic programmes could be selected, depending on resources. Within each study programme selected, students studying in a particular year could then be selected. Further, selection of a proportion of students studying in a particular year could then be made using the SRS technique. The process of selecting a sample in this manner is called *multi-stage cluster sampling*.

NON-PROBABILITY SAMPLING

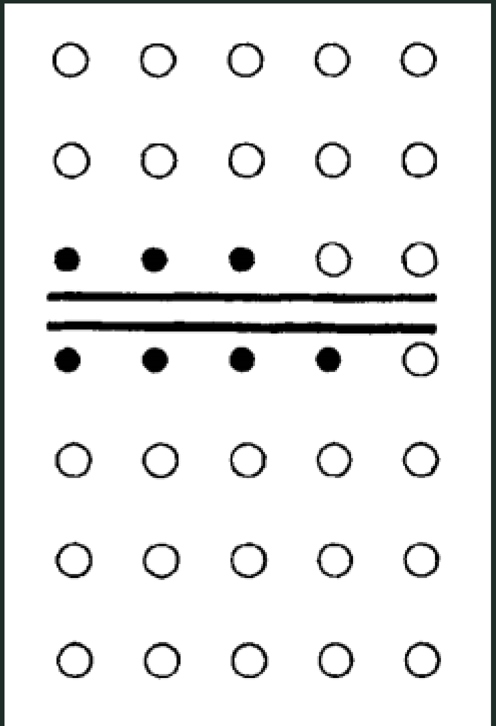
- Non-probability techniques, relying on the judgement of the researcher or on accident, cannot generally be used to make generalizations about the whole population.
- Non-probability sampling is based on selection by non-random means.
- This can be useful for certain studies, for example, for quick surveys or where it is difficult to get access to the whole population, but it provides only a weak basis for generalization.
- There is a variety of techniques that can be used, such as:
  - **accidental sampling**,
  - **quota sampling**
  - and snowball technique, etc.

QUOTA SAMPLING

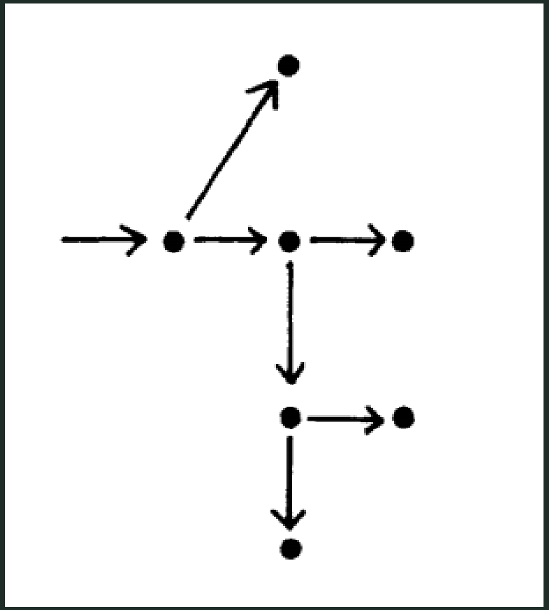
- The main consideration directing quota sampling is the researcher’s ease of access to the sample population. In addition to convenience, you are guided by some visible characteristic, such as gender or race, of the study population that is of interest to you.
- The sample is selected from a location convenient to you as a researcher, and whenever a person with this visible relevant characteristic is seen that person is asked to participate in the study.
- The process continues until you have been able to contact the required number of respondents (**quota**).
- *Example*: Let us suppose that you want to select a sample of 20 male students in order to find out the average age of the male students in your class. You decide to stand at the entrance to the classroom, as this is convenient, and whenever a male student enters the classroom, you ask his age. This process continues until you have asked 20 students their age.

ACCIDENTAL SAMPLING

- Accidental sampling is also based upon convenience in accessing the sampling population.
- Whereas quota sampling attempts to include people possessing an obvious/visible characteristic, accidental sampling makes no such attempt.
- You stop collecting data when you reach the required number of respondents you decided to have in your sample.
- This method of sampling is common among market research and newspaper reporters.
- As you are not guided by any obvious characteristics, some people contacted may not have the required information.



SNOWBALL SAMPLING



- Snowball sampling is the process of selecting a sample using networks.
- To start with, a few individuals in a group or organization are selected and the required information is collected from them.
- They are then asked to identify other people in the group or organization, and the people selected by them become a part of the sample.
- Information is collected from them, and then these people are asked to identify other members of the group and, in turn, those identified become the basis of further data collection.
- This process is continued until the required number or a **saturation point** has been reached, in terms of the information being sought.

COLLECTING PRIMARY DATA

- It needs a plan of action that identifies:
  - what data you need,
  - where the necessary data are to be found,
  - and what are the most effective an appropriate methods of collecting that data.
- There are several basic methods used to collect primary data; here are the main ones:
  - asking questions
  - conducting interviews
  - observing without getting involved
  - doing experiments

ASKING QUESTIONS

- Asking questions is an obvious method of collecting both quantitative and qualitative information from people. **Questionnaires** are a particularly suitable tool for gaining quantitative data but can also be used for qualitative data. This method of data collection is usually called a **survey**.

- *Advantages*: As a method of data collection, the questionnaire is a very **flexible** tool, that has the advantages of having a **structured format**, is easy and **convenient** for respondents, and is **cheap** and quick to administer to a **large number of cases** covering large geographical areas. There is also no personal influence of the researcher, and embarrassing questions can be asked with a fair chance of getting a true reply.

- *Disadvantages*: However, they do require a lot of time and skill to design and develop. They need to be short and simple to follow, so complex question structures are not possible.

ASKING QUESTIONS

- There are basically two question types:

1. **Closed format questions.**
    - The respondent must choose from a set of given answers. These tend to be quick to answer, easy to code and require no special writing skills from the respondent. However, they do limit the range of possible answers.
    - **Example:** Here are important qualities a prime minister should possess. Number them 1–5 in order of importance – 1 being the most important.
      Honesty Humor Intelligence Consistency Experience.
2. **Open format questions.**
    - The respondent is free to answer in their own content and style. These tend to permit freedom of expression and allow the respondents to qualify their responses.
    - This freedom leads to a lack of bias but the answers are more open to researcher interpretation. They are also more demanding and time consuming for respondent and more difficult to code.

ASKING QUESTIONS

- It is common practice to pre-test the questionnaire on a small number of people before it is used in earnest. This is called a **pilot study**.
- Questionnaires are commonly used in disciplines that are concerned with people, particularly as part of society.
- Research in social sciences, politics, business, healthcare etc. often needs to gain the opinions, feelings and reactions of a large number of people, most easily done with a survey.
- When the government wants to get answers from everyone in the population, then this kind of survey is called a **census**.

ACCOUNTS AND DIARIES

- Asking people to relate their account of a situation or getting them to keep diaries is perhaps the most open form of a questionnaire.
- These qualitative data collection methods are used to find information on people's actions and feelings by asking them to give their own interpretation, or account, of what they experience.
- You will need to transform the collected accounts into working documents that can be coded and analyzed.
- *For example*: Trainee nurses are asked to keep a diary of all their activities during a week-long work experience session in a hospital. They are asked to record what they have done in each hour of their working days and make comments on what they have learned at the end of each day.

CONDUCTING INTERVIEWS

- While questionnaire surveys are relatively easy to organize, they do have certain limitations, particularly in the lack of flexibility of response.
- Interviews are more suitable for questions that require probing to obtain adequate information.
- Three types of interview are often mentioned:
  1. **Structured interview** – standardized questions read out by the interviewer according to an interview schedule.
  2. **Unstructured interview** – a flexible format, usually based on a question guide but where the format remains the choice of the interviewer, who can allow the interview to ‘ramble’ in order to get insights into the attitudes of the interviewee.
  3. **Semi-structured interview** – one that contains structured and unstructured sections with standardized and open type questions.

CONDUCTING INTERVIEWS

- Though suitable for quantitative data collection, interviews are particularly useful when **qualitative data** are required.
- Interviews can be used for subjects, both general or specific in nature and even, with the correct preparation, for very **sensitive topics**.
- The interviewer is in a good position to judge the quality of the responses, to notice if a question has not been properly understood and to encourage the respondent to be full in his/her answers.
- Interviews can be **audio recorded** in many instances in order to retain a full, uninterpreted record of what was said. Recording and transcribing interviews do **not rely on memory** and **repeated checking** of what was said is possible. The raw data are also available for different analysis by others.

OBSERVING WITHOUT GETTING INVOLVED

- This is a method of **gathering data through observation** rather than asking questions.
- The aim is to take a detached view of the phenomena, and be ‘invisible’, either in fact or in effect (i.e. by being ignored by people or animals).
- Observation can be used for recording data about events and activities, and the nature or conditions of objects, such as buildings or artefacts.
- This type of observation is often referred to as a survey (not to be confused with a questionnaire survey) and can range from a preliminary visual survey to a detailed survey using a range of instruments for measurement.

DOING EXPERIMENTS

- Data can be collected about processes by devising experiments.
- An experiment aims **to isolate a particular event** so that it can be investigated without disturbance from its surroundings.
- They are primarily aimed at gaining data about **causes and effects** – to find out what happens if you make a change, why and when it happens and how.
- Experiments are used in many different subject areas, whether these are primarily to do with how things (objects, substances, systems, etc.) interact with each other, or **how people interact with things**, and even **how people interact with other people**.

DOING EXPERIMENTS

- It is important to check that the assumptions you make on which you base your experiment are valid.
- You can do this by introducing a **control group**, an identical setup that runs parallel to your experiment, but in which you do not manipulate the independent variables.
- A common example of this is the use of **placebo** pills for one group of patients when gauging the effect of a drug on another group. The condition of the patients taking the placebos acts as a ‘baseline’ against which to measure the effects of the drug on the condition of the other patients.

2. DATA MANAGEMENT

MANAGING YOUR DATA

- You will probably have collected a substantial amount of data for the purposes of your research project. But your data in their raw state do not constitute the results of your research.
- You would be unlikely, for example, to simply bind together transcripts of all the interviews you have undertaken, or of all the questionnaires you have had returned, or of all the notes you have taken, and present that as your report or dissertation. That would be too long and too demanding for your readers, and it would lack insight and significance.
- The business of analyzing the data you have collected, therefore, really involves two closely related processes:
  - **Managing your data**, by reducing its size and scope, so that you can report upon it adequately and usefully.
  - **Analyzing your managed set of data**, by abstracting from it and drawing attention to what you feel is of particular importance or significance.

TECHNIQUES FOR MANAGING DATA

- **Coding.**
  - The process by which items or groups of data are assigned codes.
  - These may be used to simplify and standardize the data for analytical purposes, as when characteristics like sex, marital status or occupation are replaced by numbers (e.g. replacing ‘male’ by ‘1’, ‘female’ by ‘2’).
  - Or the process may involve some reduction in the quantity of the data, as when ages, locations or attitudes are categorized into a limited number of groups, with each group then assigned its own numerical identity (e.g. categorizing ages as ‘under 21’, ‘21–64’ and ‘65 and over’, and then replacing these by ‘1’, ‘2’ and ‘3’ respectively).

TECHNIQUES FOR MANAGING DATA

- **Annotating.**
  - The process by which written (or perhaps audio or visual) material is altered by the addition of notes or comments.
  - On books or papers, annotations may take the forms of marginal notes, or (if you own the text or copy) underlining or highlighting the text itself.
  - The process may draw attention to what you consider to be the more significant sections, perhaps for later abstraction and quotation. Or it may serve as part of your continuing debate with your texts, a means to refine and progress your ideas further.

TECHNIQUES FOR MANAGING DATA

- **Labelling.**
  - Where you have an analytical scheme in mind, or are developing one, you may go through materials such as interviews or policy documents and label passages or statements with significant words (e.g. ‘mother’, ‘conservative’, ‘career break’, ‘introvert’).
  - These labels can then serve to direct your further analysis.
  - A fine distinction might be drawn between the related processes of labelling and annotation, in that labelling smacks of stereotyping, of having your ideas or prejudices worked out in advance, whereas annotating seems more open or flexible.

TECHNIQUES FOR MANAGING DATA

- **Selection.**
  - A key process in the management of data, through which interesting, significant, unusual or representative items are chosen to illustrate your arguments.
  - This may take the form, for example, of one member of a group, one institution, one answer to a survey, one particular quotation, one text, or a number of such selections.
  - The point is that you are choosing, for a variety of reasons, which examples of your data collection to emphasize and discuss.
  - There is always a good deal of subjectivity involved in such a process.

TECHNIQUES FOR MANAGING DATA

- **Summary.**
  - The process where, rather than choose one or more examples from a larger body of data, you opt to produce a reduced version, precis or synopsis of the whole data set.
  - This would probably aim to retain something of the variability of the original data collected, while saying something about the generality and/or typical cases.

3. DATA ANALYSIS

ANALYZING INTERVIEWS
Example Process 1

Thematic analysis

1. *Familiarizing yourself with your data*. Transcribing data (if necessary), reading and re-reading the data, noting down initial ideas.
2. *Generating initial codes*. Coding interesting features of the data in a systematic fashion across the entire data set, collating data relevant to each code.
3. *Searching for themes*. Collating codes into potential themes, gathering all data relevant to each potential theme.
4. *Reviewing themes*. Checking if the themes work in relation to the coded extracts (Level 1) and the entire data set (Level 2), generating a thematic 'map' of the analysis.
5. *Defining and naming themes*. Ongoing analysis to refine the specifics of each theme, and the overall story the analysis tells, generating clear definitions and names for each theme.
6. *Producing the report*. The final opportunity for analysis. Selection of vivid, compelling extract examples, final analysis of selected extracts, relating back of the analysis to the research question and literature, producing a scholarly report of the analysis.

Braun, V. and Clarke, V. (2006) Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3: 77–101

THEMATIC ANALYSIS
Example

Coding qualitative data

Interview extract	Codes
Personally, I'm not sure. I think the climate is changing, sure, but I don't know why or how. People say you should trust the experts, but who's to say they don't have their own reasons for pushing this narrative? I'm not saying they're wrong, I'm just saying there's reasons not to 100% trust them. The facts keep changing – it used to be called global warming.	• Uncertainty • Acknowledgement of climate change • Distrust of experts • Changing terminology

Turning codes into themes

Codes	Theme
• Uncertainty • Leave it to the experts • Alternative explanations	Uncertainty
• Changing terminology • Distrust of scientists • Resentment toward experts • Fear of government control	Distrust of experts
• Incorrect facts • Misunderstanding of science • Biased media sources	Misinformation

https://www.scribbr.com/methodology/thematic-analysis/

ANALYZING INTERVIEWS
Example Process 2

An orderly person spreads out her interview records in the garden

Hester was working on the records of the interviews she had carried out with a sample of students. Each record contained her typed-up shorthand notes made during the interview, and a summary of the student's background. Each consisted of several pages, including direct quotations. She first went through the interview notes, analyzing them 'question by question'. This meant having all of the records spread out at once. She wanted her analysis to be both 'professional' and 'scientific', without losing the personal touch. She preferred an orderly approach: 'I tried breaking up all of the scripts, question by question. I sat with the scripts and got out my pad, and went through each script and each question and noted down the similarities and dissimilarities. First of all I looked for common themes, and then I went through each script again noting which themes had come up'.

ANALYZING INTERVIEWS

- The first of these examples involve more experienced researchers. It, developed by qualitative psychologists, sets out a particular strategy towards thematic analysis. Like most methods of qualitative data analysis, thematic analysis works by steadily extracting from the data collected a series of themes.
- The second example involved a novice researcher who was not consciously following any particular approach to the analysis of the data she had collected. Nevertheless, the account of her analysis shows strong similarities to the other example.
- There are, of course, other approaches to the analysis of interviews. You may not produce a transcript but analyze the recordings direct. You may not have recorded the interviews but be working from your notes. …

ANALYSING OBSERVATIONS
Example

The first stage of analysis involved transcribing and importing each episode of observation into the QSR NUD*IST program. The transcriptions were read and re-read to form impressions of emerging themes and categories. A set of analytic categories were identified: inverted comma criticism, direct criticism and indirect criticism. In the second stage, data were quantified by counting instances which showed palliative care nurses doing criticism of other professionals who worked outside their organizations, for example, GPs [general practitioners] and hospital doctors. Key phrases spoken by palliative care nurses were identified. The number of times when collective pronouns 'they' or 'them' appeared in talk were counted and the number of times hospital doctors or general practitioners occurred in the nurses' talk was also counted. The constant comparative method helped to reveal systematic differences or similarities in doing criticism in each of the three palliative care settings. It also helped to identify how palliative care nurses constituted their own, their patients and other professionals' moral character. Application of the tools of CA [conversation analysis] helped to deepen analysis so as to reveal and make visible participants' local activities in palliative care nurses' talk.

Li, S. (2005) Doing criticism in 'symbiotic niceness': a study of palliative care nurses' talk. *Social Science and Medicine*, 60: 1949–59.

ANALYSING OBSERVATIONS

- A number of key points may be made about the analysis of observations in social science research:
  - Quantified forms of observation lend themselves to fairly routinized forms of data collection and analysis, which can be very powerful in getting across particular issues in tabular or diagrammatic form.
  - The collection and analysis of observation data, as with that of other research techniques, occur as much in parallel as in sequence.
  - Observation, again like other research techniques, is very often used in conjunction with other methods, both to contextualize and to extend the analysis being carried out.

ANALYZING QUESTIONNAIRES

- The data collected by questionnaires may, of course, be either qualitative or quantitative.
- Questionnaires do, however, lend themselves more to quantitative forms of analysis.
- This is partly because they are designed to collect mainly discrete items of information, either numbers or words which can be coded and represented as numbers.
- This emphasis is also partly due to the larger scale of many questionnaire surveys, and their common focus on representation, which encourages a numerical or quasi-numerical summary of the results.

ANALYZING QUESTIONNAIRES
Example Process 1

Research on livelihoods and land use patterns in southern Belize used semi-structured questionnaires to generate qualitative and quantitative data from about 100 respondents in three villages. The data was analyzed by hand because the team had no computer, and also both members of the research team could do the work together. Tables were drawn up on paper to contain the answers to each question. All the data was then entered onto these sheets and added up accordingly. Questions included, for example, enquiries about problems faced in agricultural production, producing a range of answers around: limited markets for specific products, lack of credit, and limited access to land in some places.

Laws, S., Harper, C. and Marcus, R. (2003) *Research for Development: A practical guide*. London: Sage

ANALYZING QUESTIONNAIRES
Example Process 2

A sample of 7318 rating forms from the Universidad del Pais Vasco (UPV) . . . and another sample of 90,905 rating forms from the Universidad Autonoma de Madrid (UAM) were analyzed. In both cases, students filled out a rating form [questionnaire] for each teacher from whom they received classes . . . Both questionnaires shared a focus on teacher performance in lecturing . . . The rating form applied at the UPV included 50 items. Sixteen items were dropped for these analyses, because they reflect the dimensions of fulfilment of teachers’ formal duties and exercises, as well as those items with a non-response rate higher than 10%. The overall rating items were also dropped from the analyses, as we considered that they would favour unidimensional solutions . . . Applying the same criteria, we analysed 13 of the 17 items in the UAM rating form . . . The statistical analysis was carried out by means of the structural equations model (confirmatory factor analysis, CFA) of the AMOS software and of similarity structure analysis (SSA) non-parametric multidimensional scaling.

Apodaca, P. and Grad, H. (2005) The dimensionality of student ratings of teaching: integration of uni- and multidimensional models. *Studies in Higher Education*, 30(6): 723–48.

ANALYZING QUESTIONNAIRES

- Quantitative analysis may be used, however, at a number of levels, and the simplest of these may be the most useful in your case:

Box 9.8 Levels of quantitative analysis

Descriptive statistics

Variable frequencies, averages, ranges

Inferential statistics

Assessing the significance of your data and results

Simple interrelationships

Cross-tabulation or correlation between two variables

Multivariate analysis

Studying the linkages between more than two variables

ANALYZING QUESTIONNAIRES

- Many small-scale research studies which use questionnaires as a form of data collection will not need to go beyond the use of **descriptive statistics** and the exploration of the interrelationships between pairs of variables (using, for example, cross-tabulations).
- It will be adequate to say that so many respondents (either the number or the proportion of the total) answered given questions in a certain way; and that the answers given to particular questions appear to be related.
- Such an analysis will make wide use of proportions and percentages, and of the various measures of central tendency (‘averages’) and of dispersion (‘ranges’)

ANALYZING QUESTIONNAIRES

- You may, however, wish or need to go beyond this level of analysis, and make use of **inferential statistics** or multivariate methods of analysis.
- There are dozens of inferential statistics available: three commonly used examples are: **Chi-square, Kolmogorov–Smirnov, Student's *t*-test**
- The functions of these statistics vary, but they are typically used to compare the measurements you have collected from your sample for a particular variable with another sample or a population, in order that a judgement may be made on how similar or dissimilar they are.
- It is important to note that all of these inferential statistics make certain assumptions both about the nature of your data (see Box 9.11) and about how they were collected, and should not be used if these assumptions do not hold.

ANALYZING QUESTIONNAIRES

Multivariate methods of analysis may be used to explore the interrelationships among three or more variables simultaneously. Commonly used examples of these are outlined in the table on the right.

Box 9.12 Commonly used multivariate analysis techniques

Correlation analysis

Measures the degree and direction of relationships between variables.

Regression analysis

Fits a model to a data set, enabling the prediction of the value of one (dependent) variable in terms of one or more other (independent) variables.

Analysis of variance (ANOVA)

Measures how independent variables interact with each other and impact upon the dependent variable. *Multivariate analysis of variance (MANOVA)* is used where there is more than one dependent variable.

Cluster analysis

Groups cases together into clusters on the basis of their similarity in terms of the variables measured.

Factor analysis

Reduces a large number of variables to a limited number of factors, so that the underlying relationships within the data may be more easily assessed.

Discriminant analysis

Enables discrimination between groups on the basis of predictive variables.

ANALYZING QUESTIONNAIRES

- One key point to be aware of when carrying out quantitative analyses is the question of **causality**.
- One of the purposes of analysis, we have argued, is to seek explanation and understanding. We would like to be able to say that something is so because of something else.
- However, just because two variables of which you have measurements appear to be related, this does not mean that they are.
- Statistical associations between two variables may be a matter of chance, or due to the effect of some third variable.
- In order to demonstrate causality, you also have to find, or at least suggest, a mechanism linking the variables together.

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